

Black-Scholes PDE exercise solutions

Missing formulas

[AAA]: The solution to the heat equation in the real space is given by the convolution of the $u_0(x)$ and $G(x, t)$:

$$[AAA] = \int_{-\infty}^{\infty} u_0(y) G(x - y, t) dy = \frac{1}{\sqrt{4\pi\alpha t}} \int_{-\infty}^{\infty} u_0(y) e^{-\frac{(x-y)^2}{4\alpha t}} dy$$

[BBB]: The equation U must satisfy is:

$$[BBB] \longrightarrow \frac{\partial U}{\partial t} + rx \frac{\partial U}{\partial x} + \frac{1}{2} \sigma^2 x^2 \frac{\partial^2 U}{\partial x^2} = 0$$

[CCC]: After the change of variables $\xi = \log x$ and $s = T - t$, the PDE transforms into the following:

$$[CCC] \longrightarrow \frac{\partial U}{\partial s} = (r - \frac{1}{2} \sigma^2) \frac{\partial U}{\partial \xi} + \frac{1}{2} \sigma^2 \frac{\partial^2 U}{\partial \xi^2}$$

[DDD]: The option price $C(t, x)$ is given by

$$[DDD] = \frac{e^{-r\tau}}{\sqrt{2\pi\sigma^2\tau}} \int_{-\infty}^{\infty} \psi(e^\zeta) \exp\left(-\frac{(\log x + (r - \frac{1}{2}\sigma^2)\tau - \zeta)^2}{2\sigma^2\tau}\right) d\zeta$$

[EEE]: Using the change of variables to z , we get

$$[EEE] = \frac{e^{-r\tau}}{\sqrt{2\pi}} \int_{\frac{\log K - \log x - (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}}^{\infty} e^{\log x + (r - \frac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}z} e^{-\frac{1}{2}z^2} dz - \frac{K e^{-r\tau}}{\sqrt{2\pi}} \int_{\frac{\log K - \log x - (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}}^{\infty} e^{-\frac{1}{2}z^2} dz$$

[FFF]: The first term of the Black-Scholes formula simplified into the cumulative distribution function $N(\cdot)$:

$$[FFF] = x N\left(\frac{\log(x/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}\right)$$